

— Smart Film · Trade Documentation

Installation Guide.

A step-by-step manual for installing Clearshift PDLC switchable film.
Designed so an installer doing it for the first time can deliver a
professional result.

Read this front to back before opening the roll.



Edition First print

Audience Glaziers · Tinters · Shopfitters

Pairs with Clearshift Connection Kit (CS-KIT-01)

Before you start

What this guide covers.

Clearshift PDLC switchable film is straightforward to install but unforgiving of shortcuts. This guide walks you through every step from unpacking the roll to powering on the finished panel. Read it through once before you cut anything, and keep it open beside you on the job.

If you've never installed switchable film before, that's fine — this guide assumes nothing. If you have, skip to the steps and use the call-outs as a refresher on tolerances and electrical detail.

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01 · Tools you will need.

Gather everything *before* you open the film. Starting and then realising you're missing a tool is the most common cause of botched installs.

Essentials



Isopropyl alcohol 99%

For cleaning glass and the adhesive side of the film. Never use methylated spirits.



Lint-free microfibre cloth

Two minimum. One for cleaning, one for drying. Never use paper towel.



Plastic razor blade

For glass prep only. Removes stickers, paint, dried sealant without scratching.



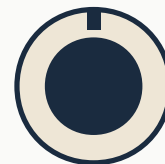
Soft-edge squeegee

Felt or soft rubber edge. Hard plastic will scratch and tear the film.



Nitrile gloves (non-powdered)

Stops finger-oil contaminating the adhesive. Powdered gloves leave residue.



Masking tape (low-tack)

For making "bunny-ear" pull tabs on the liner. Painter's tape works.



Lint roller

For picking up dust on the adhesive face before final lay-down.



Neutral-cure silicone sealant

Non-acid only. Acid-cure silicone will damage the film edges. (Included in CS-KIT-01.)



CS-CTL-60 transformer

RCM-marked 60V AC controller. Supplied with the film order.

For the electrical connection (sparky handles these)

- **16-2 stranded shielded cable** — running from controller to film, up to 30 m. Plenum-rated if running through ceiling spaces.
- **Butt-splice connectors** (red, 18 AWG) — for joining film leads to the run cable.
- **Heat shrink tubing** + heat gun — protects splices in any moisture-prone area.
- **Wire strippers, crimping tool, multimeter.**

MANDATORY — AUSTRALIA

The 240V mains connection to the transformer must be performed by a licensed electrician under AS/NZS 3000. Only the licensed electrician should make the mains-side termination. The low-voltage (60V) side of the controller and the film leads can be terminated by a competent installer.

02 · Measuring & tolerance.

Get this right and the rest of the install is easy. Get it wrong and you'll be re-ordering film.

The 3 mm rule

The cut film must be **3 mm shorter than the visible glass on every side**. That's 6 mm total off each dimension.

This gap exists for three reasons: it gives the bus bar room for the wire termination, it stops the conductive copper edge contacting the metal frame (which would short the film and burn it out), and it gives the edge sealant somewhere to sit.

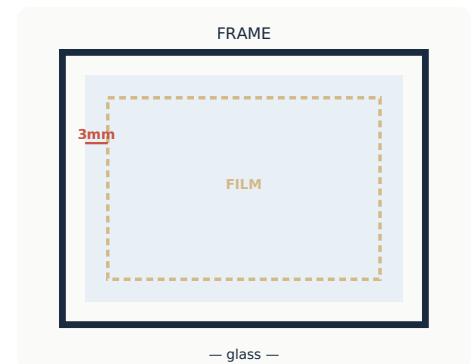
EXAMPLE Bathroom window

Visible glass area: **1200 mm × 1600 mm**

Order the film cut to: **1194 mm × 1594 mm** (6 mm off each dimension).

Measure from *visible glass edge*, not from frame to frame.

The film never sits under or behind the frame.



For frameless or beveled glass

Measure from beveled edge to beveled edge. Apply the same 3 mm setback on every side. The edge sealant will close the gap and protect the film.

For odd shapes

Sketch the shape with key dimensions and send it through with your order. We cut on a CNC plotter — anything that fits a 1200 mm width can be done.

ORDER CONFIRMATION

Every order gets a shop drawing back from us before cutting. Check it carefully.

ONCE YOU SIGN IT OFF, THE DIMENSIONS ARE LOCKED.

No on-site trimming should ever be needed.

Bus-bar position (where the wires come out)

Default position is the **top edge**, leads exiting **top-left** at 150 mm from the corner. If you need a different position (top-right, side, hidden behind a frame member), specify it on the order. Telling us after we've cut means re-ordering.

03 · Site & surface prep.

Every bubble, every speck of dust visible through finished glass started life as something you missed at the prep stage. Slow down here.

The room

- **Clean and dust-free.** Vacuum the floor and surrounding surfaces. Wipe down sills, frames and any surface within 1 m of the glass.
- **Bright, even lighting.** Bring in a worklight if needed. You need to see contamination on the glass before you trap it under film.
- **Stable temperature.** Between 15°C and 28°C. Adhesive behaves badly below 10°C and above 35°C.
- **No active dust sources.** If construction is happening nearby, isolate the room with plastic sheeting. Don't install while a wall is being sanded next door.
- **If dust is a concern** — mist the air with plain water from a spray bottle one hour before starting. This drops airborne particulates. Don't let mist hit the glass.

PRO TIP

Plan for at least two installers. Three is better for any panel over 1 m². One person handles the squeegee, others hold the film flat. Solo installation of large panels is the single biggest source of damaged film.

The glass

1. **Scrape** any stickers, paint flecks, dried sealant or putty with the *plastic* razor blade. Never use a metal blade — it pits the glass and those pits show through the finished film.
2. **Wet-clean** the glass: spray isopropyl alcohol generously and wipe with a clean microfibre cloth in one direction. Work top to bottom.
3. **Dry-buff** with a second, completely dry microfibre cloth until the glass is streak-free.
4. **Inspect** the glass from three angles — straight on, low oblique from the side, and from the room side with a torch. You're looking for anything that catches light. If you find something, repeat steps 2 and 3.
5. **Run a fingertip** (with glove) lightly across the surface. Listen and feel for catches. Catches mean residue.

✓ **DO**

- Use 99% isopropyl alcohol
- Wipe in one direction, top to bottom
- Use a fresh cloth surface each pass
- Re-check the glass under raking light

✗ **DON'T**

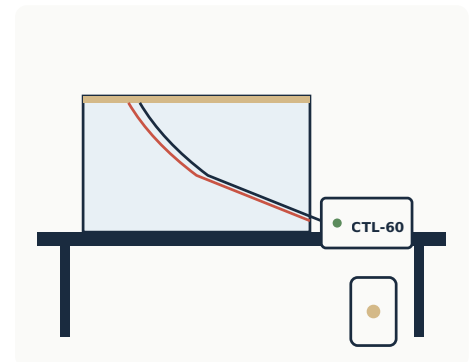
- Use water, ammonia, or methylated spirits
- Use paper towel — leaves fibres
- Touch cleaned glass with bare hands
- Install onto cold (under 10°C) glass

04 · Bench test **before** install.

Always power on the film before sticking it to the glass. Two minutes of testing now saves a refund and a re-order later.

TEST **Power the film on the bench**

1. Unroll the film onto a clean, flat, non-conductive surface (a clean trestle table covered with packing paper works).
2. Locate the two lead wires (red and black) coming from the bus bar at the edge of the film.
3. Connect the leads to the CS-CTL-60 controller — red to red, black to black.
4. Plug the controller into mains and press the remote button.
5. Film should switch from frosted to clear within 0.5 seconds.



What you're checking:

- Uniform switching across the whole sheet — no patches that stay frosted
- No dark spots, lines or streaks in either state
- Even haze level when off
- Controller LED indicators light up correctly

If the bench test fails

- **No response at all** — check controller LEDs, swap wall outlet, check leads are firmly connected.
- **Patches that stay frosted** — the film has a manufacturing defect. **Do not install.** Photograph the defective sheet powered on, email trade@clearshift.com.au within 7 days of delivery. We replace.
- **Dim or uneven switching** — usually a weak controller. Try a different CS-CTL-60 if you have a spare; otherwise contact us.

CRITICAL

Do not install a film that doesn't bench-test perfectly. Once it's stuck to the client's glass, defect replacement becomes a removal job and the warranty conversation gets harder. Always test first.

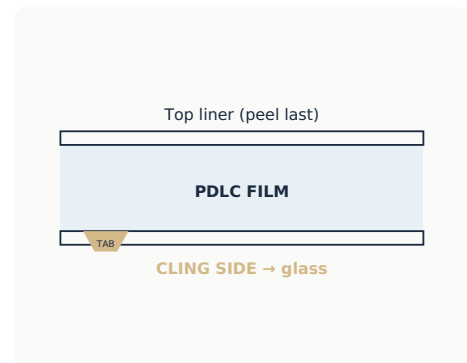
05 · Applying the film.

This is a dry installation. No water. No spray. The adhesive activates on contact with the glass — there's no repositioning once it's down properly.

STEP 1 Glove up & orient the film

Put on fresh nitrile gloves before you touch the film. Identify which side is the **cling side** (the self-adhesive face) — it's the side with the printed "CLING SIDE — REMOVE PROTECTIVE LINER BEFORE INSTALLATION" tab. This side goes against the glass.

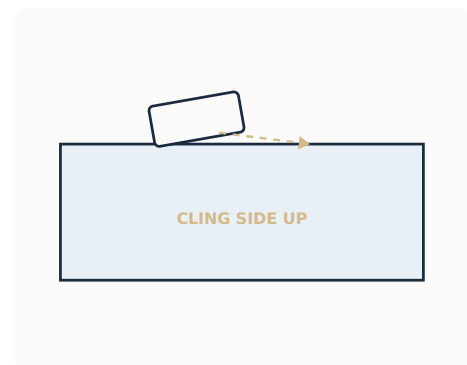
The opposite side has its own protective liner that stays on during installation and is peeled off last.



STEP 2 Wipe the cling side with alcohol

Lay the film cling-side-up on your clean work surface. Lightly mist a clean microfibre cloth with isopropyl alcohol and wipe the cling side *once* from end to end. This removes any dust picked up during transport.

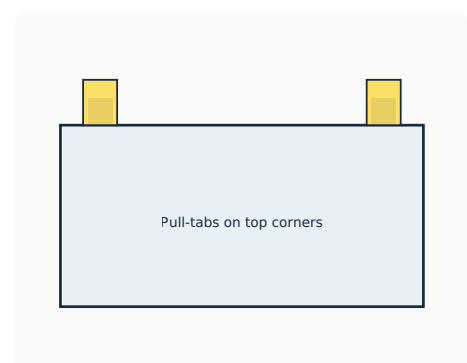
Let it flash off completely (about 60 seconds). Do not spray alcohol directly onto the cling side.



STEP 3 Make pull-tabs

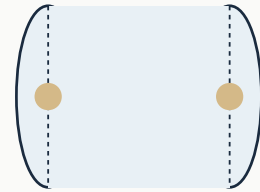
Take two strips of masking tape, about 150 mm long. Fold each into a "bunny ear" handle and stick one to each top corner of the cling-side liner. These give you something to grip without touching the adhesive.

For panels wider than 1 m, add a third tab in the middle.



STEP 4 Carry the film carefully

Roll the film loosely (do not fold or crease). Have your second installer help you carry it to the glass. **Never fold, crease or buckle the film** — internal delamination can occur and won't be visible until you've installed it.



Two-person carry · keep loose roll

STEP 5 Dry-fit and align

With the protective liner **still on the cling side**, hold the film against the glass and confirm:

- It fits with even 3 mm gaps on all four sides
- The bus bar is at the position you specified on the order (top edge, lead exit corner)
- The cling side is facing the glass

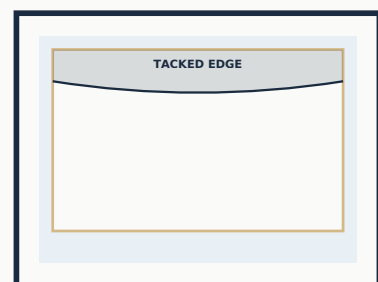
If the fit is wrong, stop. Do not proceed. Check the shop drawing.



STEP 6 Tack the top edge first

Hold the film firmly in position with the second installer. The first installer peels back the cling-side liner **only 75 mm** across the top, exposing the adhesive at the top edge.

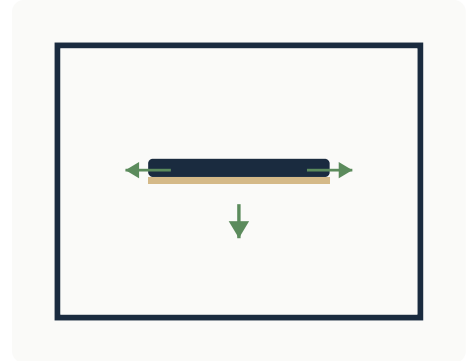
Press the exposed adhesive strip onto the glass at the top, aligning it with your 3 mm setback from the frame. Once the top edge is tacked, the film hangs in place.



STEP 7 Squeegee horizontally, work downward

Now the slow, careful work. With the top edge tacked and the second installer still holding the film flat:

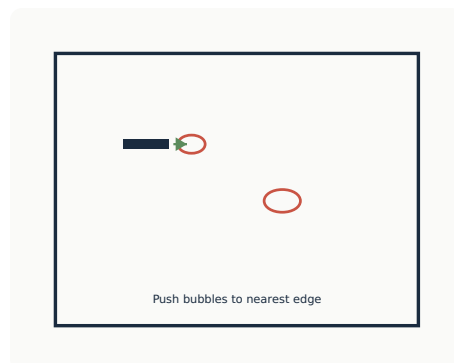
1. Peel the liner down another 150 mm.
2. Squeegee with the soft-edge squeegee from the centre outward in **horizontal strokes**, pushing air out to the sides.
3. Use moderate, even pressure. Too light leaves bubbles, too hard scratches the film.
4. Move down 150 mm at a time. Never peel more than this — you'll get caught.
5. Repeat all the way to the bottom.



STEP 8 Dealing with bubbles

Small bubbles (under 5 mm): push them out with the squeegee toward the nearest edge of the film. If they won't move, leave them — air bubbles under 3 mm typically dissipate within 48 hours as the film outgasses.

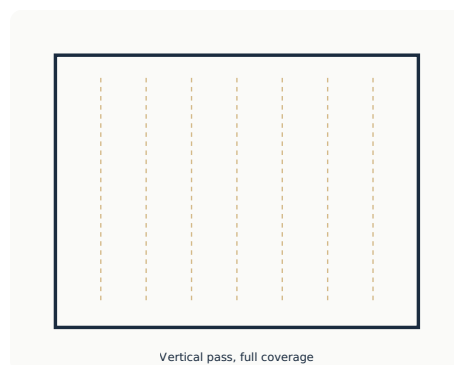
Larger bubbles or trapped debris: stop. Mark the spot with a low-tack sticker on the outside (non-cling) side. Carefully lift the film back from the nearest edge until you can access the bubble area. Use a lint roller (sticky side) on the cling face to lift off any trapped dust. Then re-lay and re-squeegee.



STEP 9 Final squeegee pass

When the whole film is down, do a complete second pass with the squeegee from top to bottom, edge to edge. Apply firm even pressure. This is what bonds the adhesive properly.

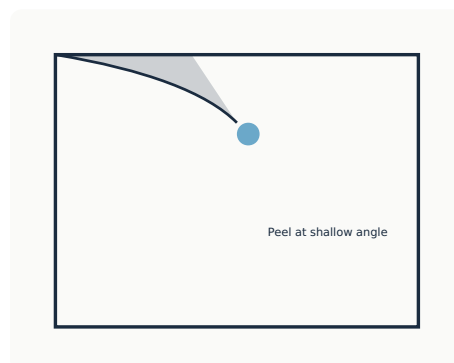
Pay extra attention to the perimeter — the last 5 mm at each edge. Lift the squeegee tip slightly so you don't run into the bus bar.



STEP 10 Peel the outer liner

The film should now be bonded to the glass. The outer (non-cling) liner is still on the room-facing side — it has its own pull tab labelled "NON-CLING SIDE — REMOVE AFTER INSTALLATION".

Lift the tab and peel slowly at a shallow angle. The liner should come away cleanly.



PRO TIP — LARGE PANELS

For anything over 2 m², it's worth bringing a third installer just for the squeegee pass. One holds the top, one holds the bottom edge of the unpeeled liner, one squeegees. Solo or two-handed installation of a 3 m² panel almost always results in a kink somewhere.

06 · Edge sealing.

The single most important step for long-term durability. Skip this and the film will start delaminating from the edges within 12 months.

Why it matters

Moisture migrating into the exposed edge of the PDLC sandwich is the leading cause of premature film failure. Sealing the edges with neutral-cure silicone locks moisture out and forms a clean, professional perimeter finish.

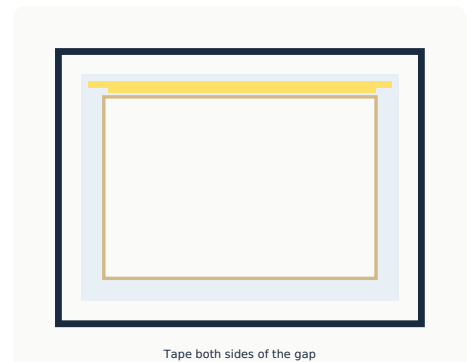
USE ONLY NEUTRAL-CURE SILICONE

Acid-cure ("vinegar-smell") silicones release acetic acid as they cure, which attacks the conductive layer in PDLC film. Check your tube — it should explicitly say "neutral cure" or "low-modulus neutral". When in doubt, use the tube supplied in the CS-KIT-01.

Step-by-step

STEP 1 Mask the perimeter

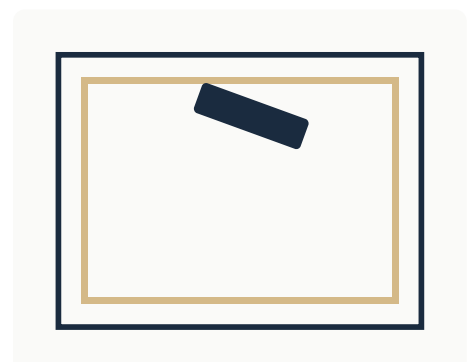
Apply low-tack masking tape parallel to the film edge, set back 4-5 mm into the gap. Mask both sides of the gap (one strip on the film, one strip on the frame or glass). This gives you a clean, straight bead line.



STEP 2 Run the bead

Cut the silicone nozzle to a 3 mm opening at 45°. Run a single continuous bead along each edge — start in one corner, work to the next, don't lift mid-bead.

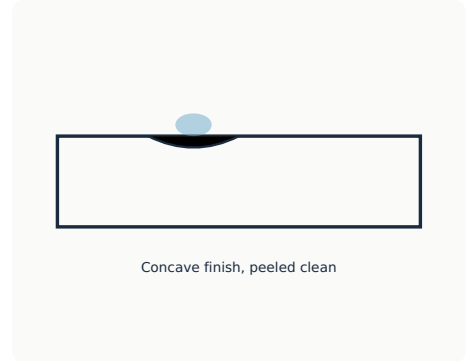
Aim for a bead that just fills the 3 mm gap. Too much and it'll squeeze out everywhere; too little and you'll leave voids.



STEP 3 Tool the bead, then peel the tape

Lightly wet a gloved fingertip with soapy water and run it along the bead at a 30-45° angle. This forces silicone deep into the joint and leaves a smooth concave finish.

Peel the masking tape immediately, before the silicone skins over. Pull at a 45° angle away from the bead. The result should be a crisp, clean edge.



Cure time

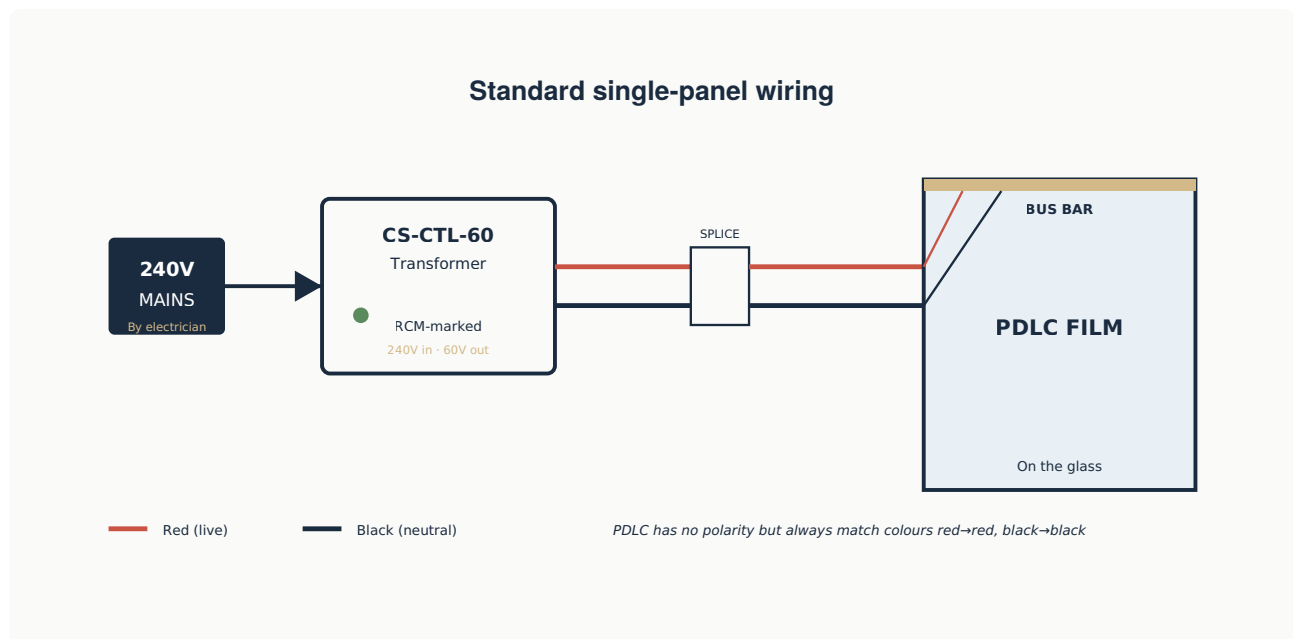
Allow **10-24 hours** for the silicone to skin and cure before activating the film. The transformer can be wired in during this time, but don't power on the film until the silicone is dry.

07 · Wiring & the transformer.

The film is an extra-low-voltage device (60V AC). The transformer is the mains-connected part. A licensed electrician handles the 240V side. The low-voltage side can be terminated by the installer.

What goes where

- **Mains 240V** → CS-CTL-60 transformer (electrician only)
- **Transformer 60V output** → run cable (16-2 shielded) → film bus bar leads
- **Optional wall switch / smart-home control** → wired into transformer's switch terminals



MULTIPLE PANELS — ALWAYS WIRE IN PARALLEL

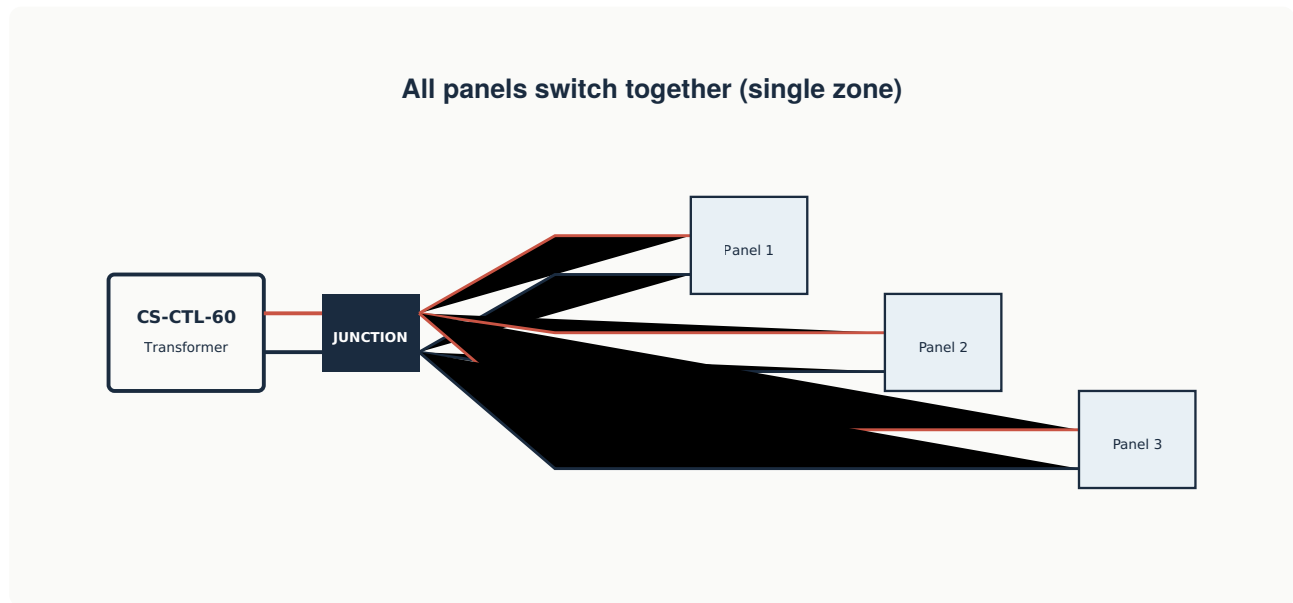
If you're running more than one panel from a single transformer, the film leads must be wired in **PARALLEL**, never in series. Series wiring will burn out the film. See the next page for parallel-wiring diagrams.

Cable specification

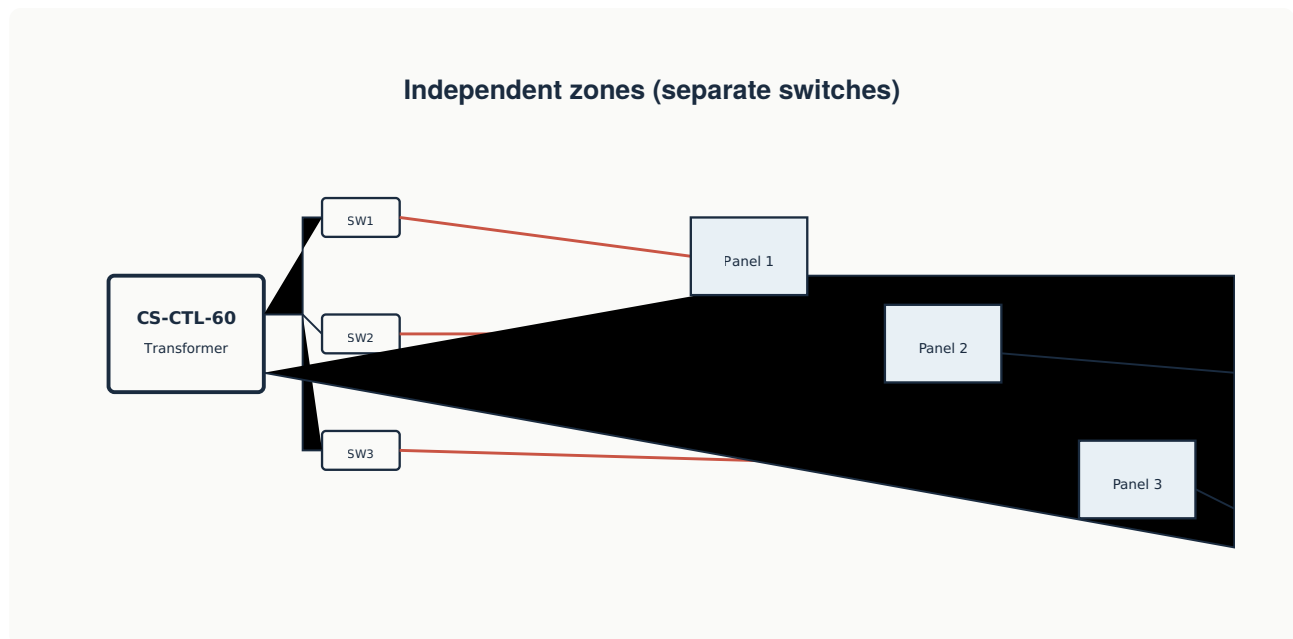
- **16-2 stranded** shielded cable for runs under 30 m. Shielding stops EMI from LED drivers and other gear interfering with the film.
- **Plenum-rated** if running through any ceiling space, fire-rated wall cavity, or air handling space.
- **One dedicated run per film panel.** Don't daisy-chain.

- Leave 1 m of slack at the transformer end for service.

Multiple panels — one switch controls all



Multiple panels — each switched independently



Making the splice

1. Strip 8-10 mm of insulation off both wire ends.
2. Insert both stripped ends into a 18 AWG butt-splice connector (red).
3. Crimp firmly with a proper crimping tool — pliers won't give enough force.

4. Slide pre-cut heat-shrink tubing over the splice (about 25 mm long).
5. Heat with a heat gun for 5-10 seconds until shrunk firmly. **Leave 25 mm of exposed wire between the bus bar and the heat-shrink** — never let shrink come into contact with the film itself.
6. Repeat for the second conductor.
7. **Red to red, black to black.** Crossing wires won't blow it up (PDLC is non-polar) but on multi-panel installs the colour discipline is what stops you from accidentally wiring in series.

08 · Power on & acclimation.

You're almost done. A few last checks and a 24-hour rest, and the film is ready to hand over.

First switch-on checklist

1. Confirm silicone has cured (10-24 hours since application).
2. Sparky has signed off on the 240V termination at the transformer.
3. All splices double-checked, no exposed copper.
4. Remote / wall switch in OFF position.
5. Switch mains on at the breaker.
6. Press ON. Film should switch from frosted to clear within 0.5 seconds.
7. Cycle the film on/off 3-5 times to confirm consistent switching.
8. Power off and leave it off for the next 24 hours.

The 24-hour acclimation

Once the film is installed and tested, **leave it switched OFF (frosted state) for at least 24 hours**. This allows:

- Any remaining microbubbles to outgas through the film (Clearshift film has a proprietary breathable adhesive that releases small bubbles over 24-72 hours).
- Edge sealant to fully cure.
- The adhesive bond to reach full strength.

THE 72-HOUR RULE

Smart film should never be left in the ON (clear) state for more than 72 hours continuously. Doing so begins to degrade the switchable interlayer. The film is designed for cyclic use — clear when needed, frosted at rest. Brief everyone who uses the room.

Final walk-through with the client

1. Demonstrate the switch — frosted to clear and back.
2. Show them the 72-hour rule and the recommended cycle.
3. Point out any small bubbles (under 3 mm) that will self-resolve over 48-72 hours, so they don't panic.
4. Hand over the warranty card.
5. Photograph the finished install — for their records and your portfolio.

MARKETING TIP

Ask the client (in writing) for permission to use the photos on your social and in your portfolio. Most will say yes. Smart film is one of those products that sells itself in a 5-second video — a frosted-to-clear toggle on Instagram is worth more than any cold call.

09 · Troubleshooting.

Most issues come from the same handful of causes. Work through the table below before calling us.

Symptom	Likely cause	Fix
Film won't switch at all	Power not reaching transformer; loose splice; failed controller.	Check breaker; check transformer LED; multimeter the 60V output. If output is dead, swap transformer.
Film switches but slow / dim	Cable run too long; cable gauge too thin; weak transformer; voltage drop at splice.	Reduce run length; upgrade to 16-2; check splice quality; try a fresh transformer.
Patches stay frosted when on	Internal film delamination, usually from folding/creasing during install OR a manufacturing defect.	Not repairable on-site. Film must be replaced. Photograph, email trade@clearshift.com.au .
Buzzing or humming sound	Transformer in a confined space; loose mounting.	Mount the transformer on a solid surface, ideally with airflow. A small amount of 50 Hz hum is normal at close range.
Bubbles after install	Dust trapped during install (small persistent), or trapped air (small self-resolving).	Mark bubble, lift edge nearest to it, lint-roll the cling face, re-lay. Bubbles under 3 mm self-resolve in 48-72 hours.
Edges lifting after weeks	Missed or incomplete edge seal; acid-cure silicone used by mistake.	Carefully scrape old sealant; re-mask; reseal with neutral-cure silicone.
Film turns "milky" in clear state	Moisture ingress through unsealed edge OR film left ON for over 72 hours.	Re-seal edges. If degradation has started from over-use, film replacement may be needed.
Flicker when LED lights are on	EMI interference from LED driver into film cable.	Use shielded cable. Route film cable away from LED driver runs. Add ferrite chokes if needed.
Remote won't pair	RF interference; remote battery flat; controller in lock mode.	Fresh CR2032 battery; re-pair per controller instructions; power-cycle the controller.
	Series wiring (not parallel); broken splice on dead panel.	Inspect wiring. Multi-panel installs must be parallel. Re-do junction if in doubt.

Symptom	Likely cause	Fix
One panel works, others don't (multi-panel)		

STILL STUCK?

Email

TRADE@CLEARSHIFT.COM.AU

with the install date, panel size, transformer serial number, photos, and a description of the symptom. We aim to respond within one business day.

10 · Care & warranty.

Brief the client on the basics. Hand them the warranty card. Send them home happy.

Cleaning the film (give this to the client)

- **Power off** the film before cleaning. Wait until it's completely dry before powering back on.
- Use a soft, lint-free microfibre cloth lightly dampened with **70% isopropyl alcohol**.
- Do not spray cleaner directly onto the film — apply to the cloth first.
- Wipe in straight, even strokes.
- Never use water, glass cleaner with ammonia, abrasive cloths, or any kind of scrubbing pad.
- Never let metal (rings, jewellery, scrapers) come into contact with the film.

The 72-hour rule (give this to the client)

Smart film is designed for cyclic use. **Never leave the film in the ON (clear) state for more than 72 hours continuously.** Doing so begins to damage the switchable interlayer. Switch it off when not in use — that's its natural resting state.

Annual maintenance (your service opportunity)

Offer the client an annual check-in for a fixed fee. In 30 minutes you can:

- Inspect wiring and splices for wear or loose connections.
- Test the transformer with a multimeter.
- Check the edge seal for any cracking or moisture ingress.
- Clean the film professionally.
- Confirm switching speed and uniformity haven't degraded.

One annual maintenance visit per install adds up fast across a portfolio.

Warranty summary

- **Film:** 5 years against manufacturing defects (delamination, switching failure, haze drift) when installed per this guide.
- **Controller / transformer:** 2 years against electrical failure.
- **Edge sealant cure:** covered if applied per these instructions with neutral-cure silicone.

- **Not covered:** physical damage, water damage from unsealed edges, mains power surges (use a surge protector), films left ON over 72 hours, installs done in dust-heavy environments, and any third-party controllers or smart-home integrations.

To make a warranty claim: photograph the film powered on and powered off, note the install date and panel dimensions, and email **trade@clearshift.com.au**. We'll respond within one business day with next steps.

REGISTER THE INSTALL

Email us the install date, client suburb (not full address), film panel size, and a photo within 14 days of completion. We log it against your trade account, which protects you on warranty claims and counts toward your Tier 2 / Tier 3 qualification.



CLEARSHIFT

Got a question? Call.

If you're mid-install and unsure, stop and call. A two-minute conversation now is cheaper than a re-order.

PHONE	04XX XXX XXX
EMAIL	trade@clearshift.com.au
HOURS	Mon–Fri · 7am – 5pm AEST
WEB	clearshift.com.au

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Smart film is a 60V AC switched electrical product. Mains connection must be performed by a licensed electrician under AS/NZS 3000.

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